

Lakshmi Bharathy Kumar

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Authorized to work in the U.S. without sponsorship

SUMMARY

Business data analyst with an M.S. in Applied Data Intelligence from SJSU (May 2026) and professional experience in web analytics and automated reporting. Focused on building interactive dashboards, automating data workflows, and delivering executive-ready insights from large and complex datasets using Python, SQL, Power BI, and Airflow.

TECHNICAL SKILLS

Languages: Python, SQL

Analytics & Modeling: EDA, Regression, Forecasting, Feature Engineering, Statistical Analysis, A/B testing, model evaluation

Visualization & BI: Power BI, DAX, Streamlit, Matplotlib, Seaborn, Excel

Automation & Engineering: Airflow, dbt, GitHub Actions, SQLite, Snowflake, Kafka, Docker, REST APIs

Databases & Tools: Pandas, NumPy, PostgreSQL, MongoDB, MySQL, Scikit-learn, Statsmodels

PROJECTS

Automated Job Monitoring & Alert Dashboard

Python, GitHub Actions, SQLite, Streamlit

- Built a zero-cost automated job tracking system using GitHub Actions scheduled workflows to scrape Greenhouse and Y Combinator listings; stores timestamped records in a SQLite database and sends real-time notifications on new role matches.
- Developed a Streamlit dashboard on top of the SQLite log to filter, sort, and visualize listings by company, role, and date — replacing entirely manual daily searches across multiple job platforms.

Price Elasticity Modeling for SKU Pricing Strategy

Python, Pandas, Statsmodels, Power BI

- Analyzed 131,706 retail transactions; filtered 5,242 SKUs down to 107 with meaningful price variation and built log-log regression models to estimate demand elasticity per SKU.
- Identified 10 statistically significant SKUs ($p\text{-value} < 0.05$, $R^2 > 0.40$); built a Power BI report presenting elasticity findings to stakeholders as direct inputs to pricing and promotion decisions.

Developer Salary Prediction & Technology Trend Analysis

Python, Pandas, scikit-learn, Power BI

- Cleaned and standardized 2020–2024 Stack Overflow Survey data (49.14% missing salary records); removed outliers using IQR and standardized experience and education fields across 90,000+ responses.
- Salary prediction model reached $R^2 = 0.6241$, RMSE \$31K; built interactive Power BI dashboards for compensation trend analysis by role, region, and technology stack for stakeholder reporting.

Demand Forecasting Pipeline with Sentiment Integration

Airflow, Snowflake, dbt, Python, Statsmodels, Apache Superset

- Designed and built an end-to-end ELT pipeline (Airflow, Snowflake, dbt) combining structured time-series data with 50,000+ social media sentiment records; engineered sentiment-weighted features for a regression model achieving $R^2 = 0.91$ at a 5-day forecast horizon; built an Apache Superset dashboard to surface forecast outputs and demand signals for stakeholder review.

PUBLICATIONS

L. B. Kumar, G. Liu, and Y. Luo, “Comparative Evaluation of Eye-Tracking Feature Representations for Gaze Direction Classification in Assistive Human–Machine Interaction,” *manuscript in preparation*, 2026.

EDUCATION

M.S., Applied Data Intelligence

May 2026

San Jose State University, San Jose, CA

- Thesis* — Evaluated 5 eye-tracking feature types and 7 model architectures for 5-class gaze classification; applied GridSearchCV, stratified cross-validation, and per-class F1 reporting — RF hit 0.90 (user-dependent), SVM/GBDT reached 0.94 cross-user, 12 pp above the published baseline.

B.E., Electronics and Communication Engineering

May 2014

Anna University, Chennai, India

EXPERIENCE

SEO Specialist — RepuNEXT, Chennai, India

Jun 2021 – Dec 2022

- Analyzed website traffic, keyword performance, and user drop-off patterns using Google Analytics and SEMrush; updated page structure and content strategy based on findings, growing organic traffic 25%.
- Built automated weekly reporting workflows for campaign metrics and audience segmentation across multiple clients, cutting manual effort and improving campaign ROI 15%.
- Translated web analytics findings into executive-ready summaries and dashboards; collaborated with content, development, and communications teams to align data insights with business strategy.